

(Q-1)

Quantum Computing and Post Quantum Cryptography.

Ans: Quantum computers can break RSA and ECC using Shor's algorithm, which efficiently solves factoring and discrete logarithm problems. To replace them post-quantum algorithms like lattice-based, hash-based and code-based cryptography are proposed. They resist quantum attacks because they rely on problems like Learning with Errors (LWE) which remain hard even for quantum computers.

(Q-2)

PRNG code in Python.

Ans: import time, os

def prng(mod=1000):

seed = int(time.time() \* 1000) + os.getpid()

return (seed \* 1103515245 + 12345) % mod

print(prng())

(Q-3)

Traditional vs Modern Symmetric Ciphers

Ans: Traditional cipher (Caesar, Vigenere) use simple substitution and are easily broken by frequency analysis.

Modern cipher like AES and DES use complex large key sizes and resist modern crypto analysis. AES is highly secure, while DES is weak today due to its short 56-bit key.

(Q-4)

Action of  $S_4$  on 2-Element Subset

Ans:  $\sigma(\{a, b\}) = \{\sigma(a), \sigma(b)\}$

This is well defined because permutations map elements uniquely.

orbit of  $\{1, 2\}$ : all 2-subset  $\rightarrow$  size

$$\binom{4}{2} = 6$$

Stabilizer: Permutations fixing the set  $\{1, 2\}$

$\rightarrow$  size 4

$$|\text{orbit}| \cdot |\text{stabilizer}| = 6 \cdot 4 = 24 = |S_4|$$

(Q-5)

Field  $\text{GF}(2^2)$  using  $x^2 + x + 1$ Ans: i) Multiplicative Group $\text{GF}(2^2)$  has elements:

$$\{0, 1, \alpha, \alpha + 1\}$$

The nonzero elements form a group under multiplication because they have inverses.

ii) ~~At~~ cyclic Nature

The nonzero set has 3 elements:

$$\{1, \alpha, \alpha + 1\}$$

A group of order 3 is always cyclic, so,  $\text{GF}(2^2)$  is cyclic.

(Q-6)

Scalar Matrices Normal Subgroup in  $\text{GL}(2, \mathbb{R})$ Ans: Scalar matrices are of the form

$$\lambda I, \lambda \neq 0$$

They form a subgroup since:

$$\text{Closed: } (\lambda I)(\mu I) = (\lambda\mu)I$$

$$\text{Inverse: } (\lambda I)^{-1} = \lambda^{-1} I$$

Normality: for an  $A \in \text{GL}(2, \mathbb{R})$ ,

$$A(\lambda I)A^{-1} = \lambda(AA^{-1}) = \lambda I$$

Scalar matrices form a normal subgroup.

Factor Group:  $GL(2, R) / \{I\}$

This is called projective linear group  $PGL(2, R)$ .

(Q-7)

Diffie-Hellman key exchange

Ans: Public: prime  $p$ , generator  $g$

Alice picks secret  $a$ , sends  $g^a \text{ mod } p$

Bob picks secret  $b$ , sends  $g^b \text{ mod } p$

Shared key:  $K = g^{ab} \text{ mod } p$

security relies on the Discrete Logarithmic Problem.

Man-in-the-middle attack: attacker intercepts and replaces keys  $\rightarrow$  solved using authentication.

(Q-8)

Intersection of Subgroups.

Let  $H_1, H_2 \leq G$

Then  $H_1 \cap H_2$  is nonempty.

If  $a, b \in H_1 \cap H_2$  then

$ab^{-1} \in H_1$  and  $ab^{-1} \in H_2$

$ab^{-1} \in H_1 \cap H_2$

Thus intersection is a subgroup.

(Q-9)

Ring  $\mathbb{Z}_n$  commutativity and zero Divisors

Ans: Addition and multiplication mod  $n$  are commutative:

$$a + b = b + a$$

$$ab = ba$$

So,  $\mathbb{Z}_n$  is commutative.

zero divisors exists if  $n$  is composite.

Example in  $\mathbb{Z}_6$ :  $2 \cdot 3 = 0 \pmod{6}$

$\mathbb{Z}_n$  is a field if  $n$  is prime.

(Q-10)

DES and AES

Ans: DES is insecure because:

→ key size only 56 bits.

→ vulnerable to brute-force search.

AES solved with:

→ Larger keys: 128 / 192 / 256 bits

→ stronger structure against cryptanalysis.

So, AES is modern secure replacement.

(Q-11)

Differential cryptanalysis resistance

i) DES Feistel structure: DES Feistel rounds provide diffusion but limited S-box size makes it vulnerable.

ii) AES Resistance: AES uses:

- SubBytes
- ShiftRows + MixColumns
- AddRoundKey

(Q-12)

Modular Inverse via Extended Euclidean Algo

Ans: If  $\gcd(a, n) = 1$  then

$$ax + ny = 1$$

$$\text{so, } a^{-1} \equiv x \pmod{n}$$

RSA uses this to compute private key:

$$d = e^{-1} \pmod{\phi(n)}$$

Efficiency is crucial since RSA uses very large primes.

(Q-13)

Block Cipher Modes

i) ECB Insecurity: ECB encrypts blocks independently:  $C_i = E_k(P_i)$

So, identical plaintext blocks  $\rightarrow$  identical ciphertext blocks.

ii) CBC Recurrence Relation

Encryption:  $C_i = E_k(P_i \oplus C_{i-1})$

Decryption:  $P_i = D_k(C_i) \oplus C_{i-1}$

(Q-14)

~~EE~~ LFSR vulnerability

Ans: LFSR are linear:

$$S_{t+k} = \sum c_i S_{t+1}$$

known plaintext allows solving the linear equations  $\rightarrow$  key recovery.

Mitigation: use non-linear combining functions or filter generators.

(Q-15)

One Time Pad :

Ans: i) Shanno Perfect Recovery Secrecy

$$P(M=m | C=c) = P(M=m)$$

Ciphertext gives no information

ii) OTP proof:

OTP encryption:  $C = M \oplus K$

with uniform random key:

$$P(C=c | M=m) = 2^{-n}$$

iii) Impracticably:

OTP requires:

→ key as long as message

→ key must be truly random

→ key cannot be reused

(Q-16)

LCG First 5 Numbers (Seed  $x_0 = 7$ )

Ans: LCG &  $x_{n+1} = (ax_n + c) \bmod m$

Example values:  $a = 5$ ,  $c = 3$ ,  $m = 16$ .

$$x_0 = 7$$

$$x_1 = (5 \cdot 7 + 3) \bmod 16 = 6$$

$$x_2 = (5 \cdot 6 + 3) \bmod 16 = 1$$

$$x_3 = (5 \cdot 1 + 3) \bmod 16 = 8$$

$$x_4 = (5 \cdot 8 + 3) \bmod 16 = 11$$

$$x_5 = (5 \cdot 11 + 3) \bmod 16 = 10$$

Five outputs are: 6, 1, 8, 11, 10

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Ring definition:

Ans: A ring is a set  $R$  with two operations

$+$  and  $\cdot$  such that:

→  $(R, +)$  is an abelian group.

→ Multiplication is associative.

→ Distributive laws hold.

Example:  $\mathbb{Z}_m$

Rings build finite fields using:  $GF(p^n) = \mathbb{Z}_p[x]/f(x)$

RSA uses ring arithmetic in  $\mathbb{Z}_n$

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RSA encrypt / decrypt ( $p=5, q=11$ )

Ans:  $n = pq = 55$

$$\phi(n) = 4 \cdot 10 = 40$$

$$e = 3$$

$$3d \equiv 1 \pmod{40}$$

$$\Rightarrow d \equiv 27$$

$$m = 2$$

Encryption:  $c = m^e \pmod{n}$

$$= 2^3 \pmod{55}$$

$$= 8$$

Decryption:  $m = c^d \pmod{n}$

$$= 8^{27} \pmod{55}$$

$$= 2$$

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RSA Signature ( $p=7, q=3$ )

$$n = 21 \quad \phi(n) = 12$$

$$e = 5, \quad 5d \equiv 1 \pmod{12}$$

$$d = 5$$

$$\text{Hash } H(m) = 3$$

$$\text{Signature: } s = H(m)^d \pmod{n}$$

$$= 3^5 \pmod{21}$$

$$= 12$$

$$\text{verify: } s^e \pmod{n}$$

$$= 12^5 \pmod{21} = 3$$

Elliptic Curve over  $P = 23$

Ans: Curve  $= y^2 = x^3 + x + 1$

i)  $P(3, 10)$

$$\text{L.H.S} = 10^2 = 100 \bmod 23 = 8$$

$$\text{R.H.S} = 3^3 + 3 + 1 = 27 + 4 \bmod 23 = 8$$

So,  $P$  lies on curve

ii) Doubling  $2P$

$$\text{Slope: } \lambda = \frac{3x^2 + 1}{2y} = \frac{3(9) + 1}{20} = \frac{28}{20} \bmod 23 = 6$$

$$x_3 = \lambda^2 - 2x = 36 - 6 = 30 \bmod 23 = 7$$

$$y_3 = \lambda(x - x_3) - y = 6(3 - 7) - 10 = -34 \bmod 23 = 12$$

$2P(7, 12)$

iii)  $P + Q$

$$\lambda = \frac{7 - 10}{9 - 3} = \frac{-3}{6} \bmod 23 = 11$$

$$x_3 = 11^2 - 3 - 9 = 109 \bmod 23 = 17$$

$$y_3 = 11(3 - 17) - 10 = -164 \bmod 23 = 20$$

$P + Q = (17, 20)$

(Q-20)

ECDSA example (mod 27,  $n=19$ )

Private key  $\Rightarrow d=9$

Public key :  $Q = dG = 9(2, 5) = (8, 12)$

Hash :  $H(M) = 8$ ,  $k=3$

$$r = (kG)_x \bmod n$$

$$kG = 3G = (5, 9)$$

$$r = 5 \bmod 19 = 5$$

$$s = k^{-1} (H + dr) \bmod n$$

$$k^{-1} = 3^{-1} \bmod 19 = 13$$

$$s = 13(8 + 9 \cdot 5) = 13(53) = 689 \bmod 19 = 5$$

Signature :  $(r, s) = (5, 5)$

verification uses:  $w = s^{-1}$ ,  $u_1 = Hw$ ,  $u_2 = rw$

(Q-21)

SHA hash Properties:

i) Secure hash characteristics

→ Preimage resistance

→ Second preimage resistance

→ Collision resistance

ii) Output length effect

SHA - 256 gives:

→ Collision security  $\approx 2^{28}$

→ Preimage security  $= 2^{256}$

iii) Applications:

→ Digital signature

→ Blockchain integrity

→ Password hashing

(Q-22)

Galois Fields in Cryptography

Ans:  $GF(p)$ : ~~integers~~ integers mod prime  $p$

$GF(2^n)$ : Polynomials mod irreducible polynomial

Used in:

→ ECC point arithmetic

→ AES uses  $GF(2^8)$  in MixColumns.

Field arithmetic ensures secure algebraic operations

i) SVP: It finds shortest nonzero lattice vector  $\rightarrow$  NP-hard  
Security basis for lattice schemes.

ii) RSA/ECC

RSA/ECC broken by shor.

Lattice problems remain hard for quantum computers.

iii) Quantum vs lattice crypto

Quantum crypto uses physics (QKD)

Lattice uses math hardness assumptions.

(Q-24)

LFSR maximum period: If characteristic polynomial is primitive, LFSR generates all nonzero states.

Number of nonzero states:  $2^m - 1$

Thus, maximum period:  $T = 2^m - 1$

### i) LWE - Based signature process

1. keyGen : choose secret vector  $s$
2. Public key : noisy equation  $As + e$
3. Sign : sample short vector  $r$  + hash
4. verify : check lattice relation holds.

### ii) Signing steps

1. Hash message  $H(m)$
2. Compute signature:  
$$z = s \cdot H(m) + \text{noise}$$
3. verify:  $Az \approx pk \cdot H(m)$